



2.4

Exponential Function Manipulation

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Exponent laws convert multiplication into addition of exponents and powers into multiplication of exponents.

2. Write one precise sentence explaining this principle: A horizontal shift can be rewritten as an output scale: $b^{x-h} = b^{-h}b^x$.

3. Write one precise sentence explaining this principle: Rational exponents represent roots and powers.

4. Write one precise sentence explaining this principle: Changing time units changes the base but not the underlying model.

Core

5. Rewrite 8^{x+1} with base 2.
6. Rewrite $5 \cdot 3^{x-2}$ as $A \cdot 3^x$.
7. A quantity triples every 8 hours. Give an hourly factor.
8. State what the formula $b^{u+v} = b^u b^v$ tells you and name any required restriction.
9. State what the formula $(b^u)^v = b^{uv}$ tells you and name any required restriction.
10. State what the formula $b^{p/q} = \sqrt[q]{b^p}$ tells you and name any required restriction.

Medium

11. Rewrite or evaluate 4^{2x-1} .
12. Rewrite or evaluate $9^{x/2}$.
13. Rewrite or evaluate 5^{x+2} .
14. Rewrite or evaluate $16^{3/4}$.

15. Rewrite or evaluate $27^{2/3}$.

Hard

16. Rewrite $12 \cdot 4^{x-3}$ as $A \cdot 4^x$.

17. Rewrite $7 \cdot 2^{x/5}$ in the form $7b^x$.

18. Show $3^{2x+1} = 3(9^x)$.

19. A model grows by factor 1.8 every 4 days. Find the 10-day factor.

Difficult

20. Find all x such that $8^{x+1} = 4^{2x-1}$.

21. Prove $b^{x-h} = b^{-h}b^x$ and interpret graphically.

22. Find q so $(81^x)^q = 3^{10x}$ for all x .

Challenge / HOT

23. Classify all transformations of b^x that can be absorbed into a multiplicative constant.

24. Find all bases c such that $2^{x/3} = c^{2x}$ for every real x .

AP-Style Multiple Choice

25. 8^x written with base 2 is
- (A) 2^{x+3}
 - (B) 2^{3x}
 - (C) 3^{2x}
 - (D) $2^{x/3}$

26. $b^{x+2} =$
- (A) $b^x + 2$
 - (B) $2b^x$
 - (C) b^2b^x
 - (D) b^{2x}
27. $16^{1/2} =$
- (A) 2
 - (B) 4
 - (C) 8
 - (D) 16
28. An h -unit growth factor for one-unit factor b is
- (A) hb
 - (B) $b + h$
 - (C) b^h
 - (D) h^b
29. $a^{m/n}$ means
- (A) $m+n$
 - (B) n th root of a^m
 - (C) a^m/a^n
 - (D) always irrational
30. Horizontal shift of a pure exponential can look like
- (A) vertical scale
 - (B) vertical shift
 - (C) reflection only
 - (D) change to linear

AP-Style Free Response

31. A quantity follows $A(t) = 600 \cdot 3^{t/8}$. (a) Give the 1-hour factor. (b) Rewrite with base e . (c) Find the factor over 20 hours. (d) Interpret.

32. Prove the equivalence $5 \cdot 2^{3x-6} = \frac{5}{64} 8^x$, then describe the transformation of 2^{3x} .

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.